VOEDING 12 V

J1 — 12V IN (GND, VIN_RAW)

Q1 — AO04405 (GS_G)

D2 — BZT52C15 (V12 → GS_S)

D1 — SMBJ15CA (V12 → GND)

C4 330u/25V · C5 330u/25V · C6 100n/25V · R4 100k · R5 100k

C1 100u/25V

BUCK 12 V -> 5 V 5 A

U1 — TPS54560B (VIN, BOOT, SW, EN, COMP, RT/CLK, GND, FB)

C7 10u/25V · C8 10u/25V · C9 10n · R6 47k · R7 100k · R8 22k · C10 100n · R9 68R

D4 — SS54 (V12 → GND)

C2 47u/50V · C3 47u/50V · C11 47u/50V · R1 10k · R2 100k · R3 10k · C12 100n · C13 100n · C14 100n

D5 — LED 5V (GND → LED5V_A) · R10 1k

J13 — 5V PI (GND, 5V)

3,3 V (ESP32 + IMU)

D6 — SS14 (V5_ESP → V5)

D7 — SS14 (V5_ESP → VBUS_ESP)

U2 — AMS1117-3.3 (V5_ESP → V3V3)

C15 10u · C16 100n · C17 100n

ESP32-S3 (SERVO-BUS + IMU + USB NAAR PI)

U3 — ESP32-S3-WROOM-1-N16R8 (3V3, EN, TXD0, RXD0, IO0, IO1, IO2, IO3, IO4, IO5, IO6, IO7, IO8, IO9, IO10, IO11, IO12, IO13, IO14, IO15, IO16, IO17, IO18, IO19, IO20, IO21, IO35, IO36, IO37, IO38, IO39, IO40, IO41, IO42, IO45, IO46, IO47, IO48, GND)

C18 100n · C19 10u · R11 10k

D8 — LED ESP (GND → LED_ESP_A)

R12 1k · R13 1k · R14 1k

SW1 — RESET (ESP_EN, GND)

SW2 — BOOT (ESP_IO0, GND)

SERVO-BUS A (L1, L2, L3)

U5 — SN74AHCT1G126 (GND, VCC, TX_A, BUSA_DRV)

U6 — SN74LVC1G34 (GND, VCC, BUSA_RXIN, RX_A)

R15 120 (BUSA_DRV) · R16 120 (TX_A) · R17 120 (BUSA_RXIN) · R18 10k (RX_A) · C20 100n · C21 100n

J2 — L1 servo (GND, 12V, BUS_A)

J3 — L2 servo (GND, 12V, BUS_A)

J4 — L3 servo (GND, 12V, BUS_A)

SERVO-BUS B (L4, L5, L6)

U7 — SN74AHCT1G126 (GND, VCC, TX_B, BUSB_DRV)

U8 — SN74LVC1G34 (GND, VCC, BUSB_RXIN, RX_B)

R19 120 (BUSB_DRV) · R20 120 (TX_B) · R21 120 (BUSB_RXIN) · R22 10k (RX_B) · C22 100n · C23 100n

J5 — L4 servo (GND, 12V, BUS_B)

J6 — L5 servo (GND, 12V, BUS_B)

J7 — L6 servo (GND, 12V, BUS_B)

IMU LSM6DS3TR-C

U4 — LSM6DS3TR-C (VDD, VDD_IO, GND, SDO/SA0, SDX, SCX, SDA, SCL, CS, INT1, INT2)

C24 100n · C25 100n · R23 4k7 (IMU_SDA) · R24 4k7 (IMU_SCL)

USB-C NAAR PI 4

J8 — USB-C (VBUS, CC1, CC2, D-, D+, SBU1, SBU2, SHIELD, GND)

J9 — USB-A PI (5V, D-, D+, GND)

R25 5k1 · R26 5k1

U9 — USBLC6-2SC6 (ESP_DP, ESP_DM, VBUS_ESP)

C26 100n

SENSOR-STROOK (G V S PER GPIO) + I2C

J14 — SENSOR S (GND, 5V, IO0–IO21)

J15 — SENSOR V (5V, IO0–IO21)

J16 — SENSOR G (GND, IO0–IO21)

J17 — (GND, 3V3, IMU_SDA, IMU_SCL)

J18 — (GND, 3V3, IMU_SDA, IMU_SCL)

R27 4k7 (IMU_SDA) · R28 4k7 (IMU_SCL)

BEVESTIGING

H1 M2.5 · H2 M2.5 · H3 M2.5 · H4 M2.5

ARACHNE-18 MINI BRAIN v1.0 — ESP32-S3 + MPU-6050 hexapod servo controller for a Raspberry Pi 4 (generated by tools/build_sch.py from tools/design.py)
Servo connectors: 1 = GND, 2 = +12 V, 3 = DATA (Feetech STS half-duplex bus, 1 Mbaud). CHECK the pin order against your cable before plugging in!
Bus A = legs L1 L2 L3 (ESP32 UART1: TX IO17, RX IO18, DE IO16). Bus B = legs L4 L5 L6 (UART2: TX IO10, RX IO11, DE IO9). DE = RTS in RS485 half-duplex mode.
Pi 4 link: USB-C (board) <-> USB-A (Pi) cable on the ESP32-S3 native USB -> /dev/ttyACM0 on the Pi. Pi VBUS only feeds the ESP32 (through D7), never the servos.
5V PI pads: buck output (5 V, 3 A for the Pi 4) -> Pi GPIO pin 2/4 (+5V) and 6 (GND), or a cut USB-C lead into the Pi power socket.
Buck = TI TPS54560B datasheet design (fig. 32): 400 kHz, 5.0 V 5 A, UVLO start 6.5 V.

ESP32-S3 + MPU-6050, 6 servo-bus legs, USB to Pi 4
Kobe Vanrobaeys – maakleerplek
Sheet: /
File: arachne_mini.kicad_sch
Title: ARACHNE-18 mini brain board
Size: A1 | Date: 2026-10-18 | Rev: 1.0
KiCad E.D.A. 10.0.3 | Id: 1/1